

Pinned group summary: GroupNamePlaceholder

Pinning-Sympy automated output

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1 Basic group attributes

Group	GroupNamePlaceholder
Matrix size	<i>MatrixSizePlaceholder</i> $\times$ <i>MatrixSizePlaceholder</i>
Rank	<i>RankPlaceholder</i>
Defining equations	DefiningEquationPlaceholder

BilinearFormPlaceholder

1.1 Lie algebra

The Lie algebra is denoted $\mathfrak{g}$. The defining equations for the $\mathfrak{g}$ are: LieAlgebraEquationPlaceholder

A generic element of $\mathfrak{g}$ is:

GenericLieAlgebraElementPlaceholder

1.2 Torus

The torus is denoted T . It has rank *RankPlaceholder*. A generic element of the T is

GenericTorusElementPlaceholder

A **character** of T is a group homomorphism from T to the multiplicative group of the ground field. The set of characters is denoted $X_*(T)$. In this document, elements of $X_*(T)$ are written as row vectors $\alpha = (a_1, a_2, \dots, a_n)$ and such a vector is viewed as a function on diagonal matrices in the following way: for $\alpha = (a_1, a_2, \dots, a_n) \in X_*(T)$ and $t = \text{diag}(t_1, t_2, \dots, t_n) \in T$,

$$\alpha(t) = \prod_{i=1}^n t_i^{a_i} = t_1^{a_1} t_2^{a_2} \cdots t_n^{a_n}$$

The rows of the matrix below are trivial characters, i.e. elements $\alpha \in X_*(T)$ which are the identity function on T . It would be more accurate to describe $X_*(T)$ as a quotient of a set of group homomorphisms modulo the lattice generated by trivial characters, but this distinction is not critical for our purposes.

TrivialCharactersPlaceholder

2 Root system

Dynkin type	DynkinTypePlaceholder
Irreducible	IrreduciblePlaceholder
Reduced	ReducedPlaceholder
Simply laced	SimplyLacedPlaceholder
Number of roots	<i>NumberOfRootsPlaceholder</i>

2.1 Dynkin diagram

The root system is irreducible and has Dynkin diagram:
DynkinDiagramPlaceholder

2.2 Cartan matrix

The root system is irreducible and has Cartan matrix:

$$CartanMatrixPlaceholder$$

2.3 Roots

Definition 2.1. A root $\alpha \in \Phi$ is **multipliable** if $2\alpha \in \Phi$.

RootTablePlaceholder

2.4 Coroots

Given a root system Φ , for each root $\alpha \in \Phi$, we find a coroot $\alpha^\vee$ (an integer vector of the same length) so that $\langle \alpha, \alpha^\vee \rangle = 2$. There is not necessarily a unique choice of $\alpha^\vee$, but for the purposes of this document, we choose $\alpha^\vee$ so that it is the integer vector of minimal Euclidean norm among those satisfying $\langle \alpha, \alpha^\vee \rangle = 2$.

CorootTablePlaceholder

2.5 Linear combinations of roots

Definition 2.2. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$.

LinearCombinationsPlaceholder

3 Root spaces

3.1 Dimensions

For a root $\alpha \in \Phi$, the dimension of the associated root space (and root subgroup) is denoted d_α . The table below lists these dimensions.

RootSpaceDimensionPlaceholder

3.2 Root space table

The table below lists each root along with a generic element of the associated root space (inside the Lie algebra). These are computed as follows: given a character α , consider the set

$$V_\alpha = \{X \in \mathfrak{g} : t \cdot X \cdot t^{-1} = \alpha(t) \cdot X, \forall t \in T\}$$

This is a vector subspace of the Lie algebra $\mathfrak{g}$. When V_α is more than just the zero vector, α is called a **root** and the root system Φ is the set of all such roots. For this document, roots are computed as follows: compile a list of potential character vectors α , which are all possible integer vectors of length n with entries from $\{0, \pm 1, \pm 2\}$, then reduce that list of potentials by quotienting by the trivial characters of the torus. Next, instantiate a generic torus element t and a generic Lie algebra element X . For each potential character α , solve the equation

$$t \cdot X \cdot t^{-1} = \alpha(t) \cdot X$$

for X . If there is a nonzero solution space, add α to the set of roots.

RootSpaceTablePlaceholder

4 Root subgroups

4.1 Root subgroup table

For the purposes of this document, root subgroups are found by simply exponentiating the associated root space. That is, for a root $\alpha \in \Phi$, the root space is

$$V_\alpha = \{X \in \mathfrak{g} : t \cdot X \cdot t^{-1} = \alpha(t) \cdot X, \forall t \in T\}$$

and the root subgroup is

$$U_\alpha = \left\{ e^X = 1 + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots : X \in V_\alpha \right\}$$

For the purposes of the groups considered in this package, the X^3 and higher terms of the exponential series always vanish, and even the quadratic term frequently vanishes, so we can simplify the description of U_α to:

$$U_\alpha = \left\{ 1 + X + \frac{X^2}{2} : X \in V_\alpha \right\}$$

The table below lists each root along with a generic element of the associated root subgroup.

RootSubgroupTablePlaceholder

4.2 Homomorphism property

For each root $\alpha \in \Phi$, the associated root subgroup map $X_\alpha : V_\alpha \rightarrow G$ is a pseudo-homomorphism with respect to the additive structure on V_α and the group operation (i.e. matrix multiplication) in G . That is, for any $u, v \in V_\alpha$,

$$X_\alpha(u) \cdot X_\alpha(v) = X_\alpha(u + v) \cdot \prod_{\substack{i > 1 \\ i\alpha \in \Phi}} X_{i\alpha}(q_\alpha^i(u, v)) \quad (1)$$

for some homogeneous polynomial maps $q_\alpha^i : V_\alpha \times V_\alpha \rightarrow V_{i\alpha}$. Thus X_α is a group homomorphism if and only if the product on the right hand side reduces to 1. The quantities $q_\alpha^i(u, v)$ are called **homomorphism defect coefficients**. For any root α of a root system Φ of any group considered in the Pinning-Sympy package, the set

$$\{i\alpha \in \Phi : i \in \mathbb{Z}_{\geq 2}\}$$

is either empty or a singleton set consisting of $\{2\alpha\}$. The non-empty case only arises when Φ is a non-reduced root system (of Dynkin type BC), and when α is a multipliable root.

Since the root system for this group is reduced, there are no multipliable roots and the product is always empty, hence each X_α is a group homomorphism there are no homomorphism defect coefficients.

5 Commutators

The commutator formula says that for two roots $\alpha, \beta \in \Phi$, the commutator of two root group elements $X_\alpha(u)$ and $X_\beta(v)$ is controlled by the structure of Φ , in particular by which positive integer linear combinations $i\alpha + j\beta$ are elements of Φ . Recall that

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

Specifically, we have the commutator formula

$$\left[X_\alpha(u), X_\beta(v) \right] = \prod_{(\alpha, \beta)} X_{i\alpha + j\beta} \left(N_{ij}^{\alpha\beta}(u, v) \right)$$

for some homogeneous polynomial functions $N_{ij}^{\alpha\beta} : V_\alpha \times V_\beta \rightarrow V_{i\alpha + j\beta}$. We repeat here the table of linear combinations of roots. The quantity $N_{ij}^{\alpha\beta}(u, v)$ is called a **commutator coefficient**. The table below gives these coefficients.

CommutatorCoefficientTablePlaceholder

5.1 Commutator identities

CommutatorEquationsPlaceholder

6 Weyl group

One of the defining characteristics of a root system Φ is that for any two roots $\alpha, \beta \in \Phi$, the reflection of β across the hyperplane perpendicular to α is another root. The reflection across the hyperplane perpendicular to α is denoted σ_α , and the previous sentence can be rephrased as saying that σ_α is a (bijective) function from Φ to itself. The reflection of β is then denoted $\sigma_\alpha(\beta)$. For each root $\alpha \in \Phi$, there is an element $w_\alpha \in G$ with the following properties:

1. $w_\alpha \in N_G(T)$, i.e. w_α normalizes the torus, i.e. $w_\alpha \cdot t \cdot w_\alpha^{-1} \in T$ for all $t \in T$.
2. $w_\alpha^2 \in Z_G(T)$, i.e. w_α^2 centralizes the torus, i.e. $w_\alpha^2 \cdot t \cdot (w_\alpha^2)^{-1} = t$ for all $t \in T$. Under some assumptions, $T = Z_G(T)$, though this is not true for all of the groups considered in this package.
3. For any root $\beta \in \Phi$, the conjugation of the root subgroup U_β by w_α is the root subgroup $U_{\sigma_\alpha(\beta)}$. That is, for $x \in U_\beta$,

$$w_\alpha \cdot x \cdot w_\alpha^{-1} \in U_{\sigma_\alpha(\beta)}$$

More precisely, given an element $x_\beta(u) \in U_\beta$, there is some function $\varphi_{\alpha\beta}(u)$ so that

$$w_\alpha \cdot x_\beta(u) \cdot w_\alpha^{-1} = x_{\sigma_\alpha(\beta)}(\varphi_{\alpha\beta}(u))$$

Taken together, properties (1) and (2) above say that viewed as an element of the quotient group $N_G(T)/Z_G(T)$, (the class of) w_α is an element of order 1 or 2.

6.1 Weyl group elements

The table below lists roots along with an associated Weyl group element.

WeylGroupPlaceholder

6.2 Weyl group conjugation coefficients

The table below lists roots along with the coefficients obtain in performing conjugation by an associated Weyl element.

WeylConjugationCoefficientPlaceholder

6.3 Weyl group conjugation identities

WeylConjugationEquationsPlaceholder